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Generation of fashionable clothes using generative adversarial networks

A preliminary feasibility study

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Abstract

Purpose – There are various style options available when one buys clothes on online shopping websites, however the availability the new fashion trends or choices require further user interaction in generating fashionable clothes. The paper aims to discuss this issue.

Design/methodology/approach – Based on generative adversarial networks (GANs) from the deep learning paradigm, here the authors suggest model system that will take the latest fashion trends and the clothes bought by users as input and generate new clothes. The new set of clothes will be based on trending fashion but at the same time will have attributes of clothes where were bought by the consumer earlier.

Findings – In the proposed machine learning based approach, the clothes generated by the system will personalized for different types of consumers. This will help the manufacturing companies to come up with the designs, which will directly target the customer.

Research limitations/implications – The biggest limitation of the collected data set is that the clothes in the two domains do not belong to a specific category. For instance the vintage clothes data set has coats, dresses, skirts, etc. These different types of clothes are not segregated. Also there is no restriction on the number of images of each type of cloth. There can many images of dresses and only a few for the coats. This can affect the end results. The aim of the paper was to find whether new and desirable clothes can be created from two different domains or not. Analyzing the impact of "the number of images for each class of cloth" is something which is aim to work in future.

Practical implications – The authors believe such personalized experience can increase the sales of fashion stores and here provide the feasibility of such a clothes generation system.

Originality/value – Applying GANs from the deep learning models for generating fashionable clothes.

Keywords Neural network, Cross-domain relations, Fashion clothes, Generative adversarial networks

Paper type Conceptual paper

1. Introduction

Fashion is a constant presence in a person's life but it is not consistent. People follow fashion to express themselves. One combines his clothes according to his own opinion and creates a unique style. As fashion trends change so frequently, some of them are very extreme and opposite to what a person likes. Recent advances in the artificial intelligence domain and especially in machine learning has led to tremendous collaborative works between the fashion and computer-based designs. In the textile and fashion world cloth generation systems that

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take images of latest fashion trend and the clothes purchased by the customers as input to produce new line of trends are highly desirable. In this work, we study a deep learning based system, namely, generative adversarial networks (GANs) that utilizes the input information to generate new clothes, which are trendy and at the same time are derived from the user style. These can be also be used as a blueprint design by the manufacturers to make new clothes. Such a system will give a personalized shopping experience to the person.

Generation of new and trendy clothes based on based on the current fashion trend by understanding the styling of the customer using GANs a new topic and to the best of our knowledge it has not been studied in this context. To generate such type of data the system must be able to relate the two different image domains. For this purpose, we need to create function, which maps images from one domain to the other, thus identifying the cross-domain relationships. This can be achieved using a GAN couple. This system will be supported by a relevance feedback in which the output is customized based on previous selections made by the user. If the user selects a particular image, then the weight all the images that have the same style as the selected image will be increased. This weight will be used to sort the generated images in order to their relevance to the user. We believe our system can have many applications. The same system can be employed to create new fashion clothes from old fashion clothes. This will need two types of images – one will be from latest fashion and the other will be from old fashion. This system can also be modified to work for clothes for different occasions as well. For this purpose, the data set needs to be updated accordingly. Such a system can also change the process of designing and manufacturing clothes. The designers can understand what people like and can use their creativity efficiently.

Typically, GANs require a considerable amount pre-labeled data for the training purpose. The images must have the similar size with minimal noise. However, to generate clothes which are based on the consumer style we need thousands of cloth images for each user, if personalize the system for each user. This is not possible in real life as a person might never purchase so many clothes. Nevertheless, there can be some other person living far away from our first user, which has the same dressing style. If we are somehow able to find all the users with similar dressing sense, then we can solve this problem. This can be achieved using similarity techniques. First, we will find color and texture features. These include color auto correlogram, color moments, Gabor wavelet transform and HSV histogram. This will give us a feature vector on which we can apply clustering techniques to form clusters of similar dressing styles. Unfortunately, the requirements for such a data set cannot be fulfilled by any existing fashion data sets. What is even more difficult is to find the data set, which includes clothes, liked by a particular user. Creating new fashion clothes from old fashion clothes is also related to the identification of the cross-domain relationships. By identifying cross-domain relationships with a GAN variant (Kim *et al.*, 2017), we condition one set of images based on the other. In other words, we try to generate new images from a domain which will try to match some features of the other domain. Therefore, both problems have a common solution. If the system works well for the latter, it will surely work for the former data set. In this research, we work on solving the second problem. Rest of this work is organized as follows. Section 2 details the related works, and Section 3 introduces our GAN based fashionable clothes generation. Section 4 provides our experimental results and Section 5 concludes the paper with discussions.

2. Related works

A GAN includes two interconnected systems. One system tries to learn patterns from the original data so that it can discriminate between what is original and what is fake, when a data item is presented to it after training. Therefore, this system can be thought of as a discriminator. The second network will try to generate the data which it similar to the original data, starting from some noise. This system is called a generator. The generated data will then be passed to the discriminator for the evaluation. The discrimination error from this system will be used by the

generator to improve. Therefore, such a model forms the base of our system. However, the data generated in such a manner will be of the type on which the discriminator was trained.

GANs have been used for previously (Zhu *et al.*, 2017) to generate new clothes based on the description provided by the user and at the same visualize himself in those clothes. This research develops a system capable of learning the posture of the person and then dresses the person with this newly developed dress by GAN. This shows that GANs are capable enough to learn the complex clothing patterns. Nevertheless, the newly generated clothes will similar to what is in the market which means the latest trend or the data set used for the learning purpose. This means the system is capable of capturing a one domain or type of clothes. What we aim is to generate new clothes, capturing the ideas of two very different domains.

Some researchers (Matzen *et al.*, 2017) have tried to understand the most popular styles in different cities of the various countries around the globe. Therefore, people in different cities follow different fashion trends. This can form the third type of problem where the first domain includes fashionable and latest clothes and the second domain includes clothes popular in that region. The data set used by the researchers (Matzen *et al.*, 2017) is based on the pictures taken from the social networking websites like Instagram. Such images are very noisy and are not consistent in nature. By noise and consistency, we mean that the background of the images is not same, lighting and the capturing angle is different in every picture and the picture captures different poses. This makes the learning process extremely difficult.

There are many existing fashion based recommendation system that will recommend users new dresses that the user might like based on the previous shopping history of the user. The recommended item is similar to or complements the items that have already been purchased by the user or the items that the user is interested in (Gu, Liu, Cai and Shen, 2017). But these recommendation systems recommend already existing fashion styles or designs that are there in the database. This is where the novelty of our system comes in to generate new designs and patterns based on the trendiest items that are in fashion on a single piece of cloth.

There are some existing systems that will generate new images for people in different poses based on the image of a person wearing clothes (Ma *et al.*, 2017). These types of systems employ the technique of pose integration and image refinement based on the existing image to synthesize the image of the same cloth when looked from a different angle. So, no new designs are generated in these systems.

Few existing systems such as Liu *et al.* (2016) can analyze the styles of the clothing that the user is interested in based on the feature extractions of the clothes such as visual features, distribution of styles, aesthetic word spaces, etc. They also combine the features of tops and bottoms. Some systems like Chen *et al.* (2015) extract the visual features such as the colors and the designs of the clothes to find out the similarities in the styles of different clothes. Chen *et al.* (2015) constructs a database of the clothes from the New York Fashion Shows and compare the trends. They elaborate on the current trends on fashion and changes in styles. But no new patterns or clothes are generated in these systems either. These modern styles and trends from different fashion shows have been incorporated in our system too.

There are few systems like Gu, Wong, Peng, Shou, Chen and Kankanhalli (2017) which take into account the street fashion data and identify some of the long term and short term fashion styles. We used the styles which are long-term and have a perennial presence in the fashion industry for the generation of our data set so that the new images generated by our system will also include the patterns that are always in trend.

Some proposed systems like He *et al.* (2016) check the similarity of trends and the pattern of the items that the user is interested in. These systems will not only compare clothes, they can also compare other items like shoes, jewelry, handbags, etc. These systems help us in gaining valuable insights into fashion and the current trends.

Few models like Lassner *et al.* (2017) a semantic segmentation of the body and clothing is generated and then a conditional model is learned on the resulting segments that creates

realistic images. But they are taking into account, the entire body structure, due to that the design generations and the details of the patterns are not clear, and hence they cannot be used by the manufacturers to make new clothes. Our system on other hand takes the cloth piece alone and generates new and clear design patterns.

Some other systems like Xian *et al.* (2018) use the textures given by the user to synthesize objects consistent with the texture suggestions. They can also be applied on a wide variety of objects such as clothes, shoes, purses, etc. The details of the textures can be controlled using texture patches. Though this system will give user the ability to customize, but from the millions of designs, it is not possible to for the user to generate the best design or to search for the trendiest texture patch. This also poses a problem for cloth manufacturers as there is no parameter to decide which texture they should use for the generation of designs that will be in fashion for the longest time so that those designs can be used for mass production.

3. Our proposed GAN based model

GAN architectures like image-to-image translation (Najafabadi *et al.*, 2015), take image from one domain and generates an image in other domain. But when we want to establish relationships between the two domains, pairing of images is required which is very labor intensive. DISCO – GAN (Kim *et al.*, 2017) architecture solves this problem of finding cross-domain relationships by using two different GANs.

For instance we have two domains one of handbags and other of shoes. To generate a shoe image based on each handbag, we need to find cross-domain relationships between them. This will help to generate an image which likes in the domain of shoes but at the same time is a representation of that handbag image. To check if the image lies in a particular domain or not, we use a discriminator and to force the generated image have elements of the original domain mode we use a reconstruction loss. Therefore, one-to-one mapping is developed.

To generate image of other domain from a given domain while keeping its features intact can be done by utilizing cross-domain relationships (Kim *et al.*, 2017). Consider two domains A and B.

F_{AB} is a function that converts an image of Domain A to Domain B. Similarly, F_{BA} is a function that converts an image of Domain B to Domain A. Therefore cross-domain relations can be described as a function that maps Domain A to B and vice versa.

D_B is the discriminator that distinguishes the generated images from the domain B. Similarly, D_A is the discriminator that distinguishes the generated images from the domain A. F_{AB} and F_{BA} are the generators.

In a traditional GAN the generator tries to generate image of one domain by using error from the discriminator. Rather than using a single generator, the system uses two generators in such a way that the first generator will be F_{AB} and the second generator is F_{BA}.

If x is an image from domain A and $x1$ is an image from domain B. F_{AB} will generate an image of domain B. This generated image is y. y will be given to F_{BA} to generate image z. z should be from domain A. The distance between z and x can be defined as reconstruction loss RECON_A. This distance can be measured by L1, L2 or Huber Loss. The system tries to minimize this distance. y is given to D_B to get the ERROR_B:

$$\text{ERROR}_B = \log D_B(x1) \log(1 - D_B(F_{AB}(x))).$$

The error given to the F_{AB} we be the error encountered by the discriminator D_B (ERROR_B) and the RECON_A:

$$\text{Loss}_{AB} = \text{ERROR}_B + \text{RECON}_A. \tag{1}$$

This loss is a combination of two losses. The first one will try to convert image of domain A to domain B and the second loss tries to retain the properties the domain A. Therefore such a system is successful in mapping Domain A to B.

But this forms one direction mapping. For the bijective function we need to convert Domain B to A. The similar equation loss can be developed for F_{BA} as well:

$$\text{Loss}_{BA} = \text{ERROR}_{\mathbf{A}} + \text{RECON}_{B}. \tag{2}$$

where ERROR_B is the error from D_B and RECON_A reconstruction distance loss between Domain B and the image finally generated from F_{AB}.

The total loss is defined as $\text{Loss}_{AB} + \text{Loss}_{BA}$. This system is now capable of identify cross-domain relations. The architecture diagram below has describes the same idea (Figure 1).

3.1 Optimization

The DISCO-GAN (Kim *et al.*, 2017) architecture uses two GANs, which makes the system very computationally expensive. Therefore, any optimization can reduce the training time. The original architecture uses the standard convolutional operations.

For instance, we want to apply a kernel of size $(D_K \times D_K)$ on an image of size $(D_F \times D_F \times M)$. Where M is the number of channels. The size of the image after convolution is $(D_G \times D_G)$. The total number of operations performed if, the output has N channels are:

$$N \times D_G^2 \times D_K^2 \times M.$$

The number of computations can be reduces using depthwise separable convolutions (Isola *et al.*, 2017; Chollet, 2017):

(1) Depthwise convolution.

We apply M filters of size $(D_K \times D_K \times 1)$ to each of the M channels of input image of size $(D_F \times D_F \times M)$. We get an image $(D_G \times D_G \times M)$. The total number of operations:

$$M \times D_G^2 \times D_K^2.$$

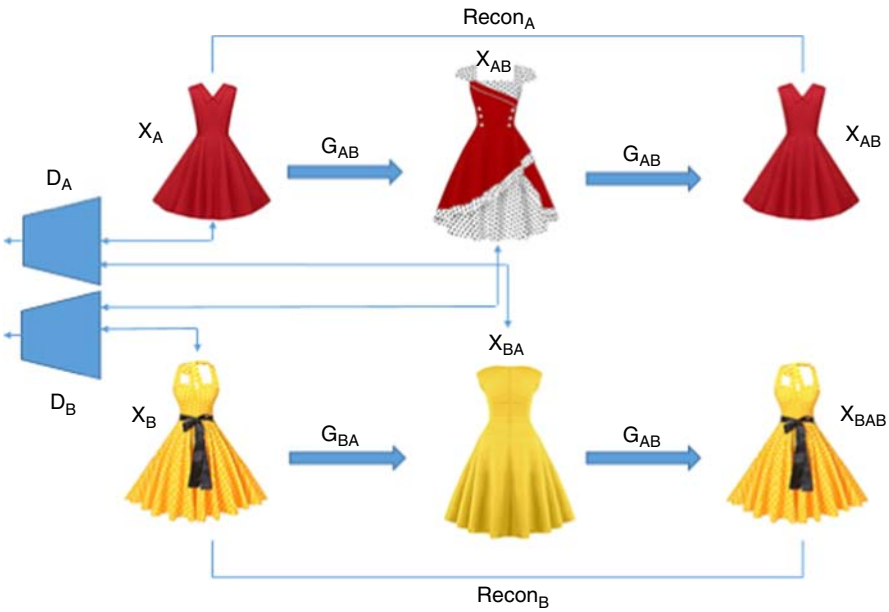

Figure 1. Representations of the proposed architecture of the GAN-based system

(2) Pointwise Convolution.

We apply N filters of size $(1 \times 1 \times M)$ to get an out of $(D_G \times D_G \times N)$. The total number of operations:

$$N \times D_G \times D_G \times M.$$

$$\text{The total number of operations} = M \times D_G^2 \times D_K^2 + N \times D_G^2 \times M$$
$$= M \times D_G^2 \left(D_K^2 + N\right).$$

$$\frac{\text{Number of operations in depthwise separable convolutions}}{\text{Number of operations in standard convultions}} = \frac{M \times D_G^2 \left(D_K^2 + N\right)}{N \times D_G^2 \times D_K^2 \times M}$$
$$= \frac{1}{N} + \frac{1}{{D_K}^2}.$$

If $N = 1{,}024$, $D_K = 3$, the above ratio is 0.112. This means that number of operations in standard convolution are 9.28 times the number of operations in depthwise separable convolutions. The same is the result for the ratio of number of parameters. Therefore, this approach can strongly reduce the training time.

3.2 Relevance feedback

Customization of results based on the user feedback is very important for a system which gives fashionable clothes as an output and primarily uses the likeness by the user as a parameter for success to fulfill its goal.

For relevance feedback, initially, the system will show the user the results in the order in which they are generated. After that, the user will select the dresses that he/she likes. The selected dresses will be considered as the optimum dresses as far as the design, the patterns and the colors are concerned. For the next iteration, the selected dresses will be shown at the top and the dresses that are similar to the selected dresses in design and patterns will be shown after that. The dresses generated by the system that differ a lot in designs and colors from the user selected dresses will be shown at the last. Relevance feedback in such systems can be implemented in two ways:

(1) One way is that the dresses that are selected are the only correct pattern, i.e. the user likes only the selected dresses, so all the other dresses will be considered as an irrelevant choice and they all will be shown at the last.

(2) The second way is that although the user has not selected a particular dress that can also be useful to the user since maybe the user has not seen it. It is possible as the system will generate thousands of images and it is not practically feasible for the user to look at each and every image and classify it into the classes whether the user likes it or not.

The second option is more feasible initially when the user is starting with the system and he/she has not made any purchase or classified any dress into useful or not. So, in that case to show the variety of results in the absence of any constraint of what type of dresses the user likes, the second option performs the best.

Once the user has enough experience with the system, he/she has made enough purchases using the system (number of purchases are above a certain threshold A) and the user has classified enough images into the categories whether he/she likes the dress or not (number of

classified dresses are above a certain threshold B), i.e. the system has enough information to see the pattern in the kind of dresses that the user likes and is efficiently able to predict whether the user will like a new dress or not, then the system will switch from the second type of relevance feedback to the first type of relevance feedback to give the user a more customized and seamless experience. This switch is made to give the user a better user experience since there is no need to show the dresses the user does not like and will not purchase. So, the system will adapt to show only the types of dresses the dresses that the user will like and hence purchase.

4. Experimental results

4.1 Data set

The system aims to develop dresses which are fashionable and at the same time has some features from the vintage dresses. To develop such a system, we need two types of data sets. The first type of data set includes Fashionable dresses and the second type of data set includes vintage dresses. Unfortunately, there is no pre-existing data set that can be used. To solve this problem, we used a custom-script written Image Crawler to download and aggregate images of dresses data set from internet with copyright free sources. GANs are typically data expensive models and require large number of training images. To train we require a minimum of 50,000 images per domain. To match this training data requirement, we applied some image data set augmentation techniques. In total, 6,000 images were collected for each domain. Then, using the following image augmentation techniques, the size of the data set was increased to 80,000 for each domain. For this type of problem, applying multiple HUEs will represent dresses in different colors. Such a technique will be suitable for the image augmentation task because, it directly conveys the idea of having the same dress in different colors. Since the data set was collected automatically, there are no set rules that are followed. In general, it is preferred to have clothes images that have a uniform white background and no other object other than cloth itself. This means that the data set has minimal amount of noise, so no specific changes to the data set or the GAN model were required.

Technique1 – six different HUEs were applied to the images of each domain separately to increase the size of the data set. Given the HUE scale from −180 to 180, the following HUE operations were applied 180, −180, 90, −90, 45, −45. By changing the HUE of the images, the color of the dress changes. As a result we get one dress in six different colors plus the original color. This gave us 42,000 (36,000+6,000) images.

Technique2 – each image was further divided into four parts, where each part of the image was given a different HUE. The following HUEs were applied sequentially 30,−30,120 and −150. The result of this operation gave us $42{,}000 \times 2 = 84{,}000$ images.

Exact same operations were applied for both the domains. The images collected have no background and humans. This allows the system to converge quickly and find the relations between the clothing types only.

4.2 Results

4.2.1 Expected outputs. Data sets under consideration.

Domain A – old fashion clothes (Figure 2).

Domain B – new fashion clothes (Figure 3).

Expected output – generated clothes should have features of new as well as old-fashioned clothes.

The results obtained show the transition from Domain A to new generated image in Domain B. From this generated image to Domain A. For the initial epochs the output majorly consists of noise. After running the system for about 5,000 epochs, one will get good outputs.

4.2.2 Actual outputs. Table I.

Figure 2.
Sample from an old fashion clothes data set

Figure 3.
Sample from a new fashion clothes data set

Old Fashion Clothes	New Fashion Generated Clothes

Table I. Examples of old fashion clothes that are used to generate new fashion clothes

5. Discussion and conclusion

Table I shows the examples obtained from the GAN approach. The images generated above are of 128 × 128 resolution, and the new fashion generated clothes have attributes of the old fashion clothes. In the first pair, the net structure of the gown has been transferred to a full

sleeve t-shirt. In the second pair, the small printed blocks have been transferred to the blouse by keeping color combinations in mind. The blue colored blocks are transferred to the region where the background is also a shade of blue. In the third pair, the polka dot design has been transferred along with the parts of mustard colored flower. In the fourth pair the ducks have been transferred to the new top. In the fifth pair the overall color of the cloth has been changes to match the design of the original old fashion cloth. As developed system works on the concept of GANs. Therefore, the loss and efficiency is measured using the visual perceptions of the newly formed design patterns. The main drawback of the system is, it is highly depend on the number of training images. With very less data the system will not be able to learn, therefore the size of each domain must be around 80,000+.

The biggest limitation of the collected data set is that the clothes in the two domains do not belong to a specific category. For instance the vintage clothes data set has coats, dresses, skirts, etc. These different types of clothes are not segregated. Also there is no restriction on the number of images of each type of cloth. There can many images of dresses and only a few for the coats. This can affect the end results. The aim of the paper was to find whether new and desirable clothes can be created from two different domains or not. Analyzing the impact of "the number of images for each class of cloth" is something which is aim to work in future.

To conclude, we were able to create our own virtual cloth generation consisting of new designs and patterns with GANs. These newly created designs have attributes of both old and new fashions clothes. If the same system is trained for clothes bought by the user then we will get clothes, which are trendy as well as personalized for different consumer types. But the performance of this system is highly dependent of the size of the data set. The user has no control over the output being generated. One does not know *a priori* which two images will pair up and what exactly will be the design on the new generated cloth. In future one can try to modify the system in which the users express what the wants from the system, especially with mobile app based vision system (Howard *et al.*, 2017).

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